MODERN WEB ENGINEERING: 8-WEEK INTENSIVE COUR

WEEK 2: COMPONENT ARCHITECTURE, REACTIVE STATE & HIGH-PERFORMAN

Official Student Study Guide & Reference Materials

Confidential - Authorized Enrolled Student Access Only

MODULE OVERVIEW:

Design composable component hierarchies, understand virtual DOM diffing, master custo
m state hooks, and optimize rendering pipelines.

KEY LEARNING OBJECTIVES & CORE SYLLABUS:

[01] Atomic Design & Modular Component Hierarchies

[02] Reactivity Models & Virtual DOM Reconciliation

[03] Custom Hook Factories & Fine-Grained State Primitives

[04] Render Tree Profiling, Memoization & Layout Shifts

PRACTICAL EXERCISES & LAB MILESTONES:

- Complete the hands-on code exercises described in repository module 2.
- Submit your architecture review via the designated student portal or WhatsApp advisory.
- Verify test coverage and pass all CI assertions before progressing to the next module.

Instructor Support: Contact Course Administrator via WhatsApp for 1-on-1 feedback.

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